Data dictionary and verification guide

The Heat to Come · Application 2.5.0 · Model V2-2.3.0 · G2 dossier 2026-09-26

Audience: facilitators, analysts and technical reviewers. Language: English. Intended destination: public technical appendix. Editable master: Markdown, with PDF/DOCX delivery copies. Technical data extract: `technical-data-extract.json`, a parameter companion, not an importable player save or a replacement engine.

File roles and versions

The extract has schema `g2-technical-extract-v1`, application version 2.5.0 and model version V2-2.3.0. It contains the current parameters, four scenarios, 36 interventions with 108 level rows, 272 interactions, six concentration rules and 48 events. The extract preserves English event modelling rationales. Italian internal pair/concentration notes are omitted; numeric records are unchanged. It is an audit extract created for this dossier, not a canonical runtime source. Editing it does not change the game.

Canonical sources remain `data/model/v2_3_model.json`, `data/events/v2_events.json` and `data/content/v2_3_en-GB.json`, all relative to the private V2 source root. The engine is `app/src/domain/engine-v2-3.mjs`. Event schema 2.2.0 and engine schema 2.3.0 coexist intentionally in application 2.5.0. The released application's compatibility contract requires the corresponding complete fingerprint set; matching a version string alone is insufficient.

Model fields

Field / selector	Type and units	Meaning
scenarios.PC / FF / CZ / SR	Object	Four stable short scenario keys
scenarios.*.runtimeId	String	Stable application slug
riskBase / resilienceBase	Index points	Initial synthetic metrics
budget	Relative resource units	Nominal three-turn programme budget
releases	Three numbers	Exact scheduled releases; sum equals budget
floor	Risk index points	Scenario minimum used by risk updates
readiness.*	Multiplier	Category conversion of nominal risk and resilience
weights.*	Percentage-point weights	Five weights summing to 100
interventions[].runtimeId	String	Stable INF/OPS/PRE/GOV/BEH/RES identifier
interventions[].id	String	Stable INT-V2 record identifier
levels[].level	Integer 0, 1, 2	Not selected; limited/pilot; comprehensive
levels[].cost	Relative resource units	Nominal capital cost at the level
levels[].risk	Reduction-strength points	Input to risk asymptote, not direct risk subtraction
levels[].resilience	Signed contribution	Nominal positive or adverse resilience effect
levels[].acceptability	Index points	Input to robust package acceptability
lifecycle.class	Classification string	permanent / persistent / temporary; no expiry timer
lifecycle.activation.l1 / l2	Turns	Activation delay from the operation
lifecycle.maturityCurve	Three coefficients per path	Physical benefit delivery over operation year and two following years
lifecycle.tags	String array	Capability tags used by events and coordination

Field / selector	Type and units	Meaning
lifecycle.repairCost	Relative resource units	Contextual repair price, not annual upkeep
falseFriend	Boolean	Authored maladaptation flag; excluded from capability contribution
deliveryClass	String	immediate / programme / infrastructure
interactions[].id1 / id2	Stable IDs	Unordered authored pair
r_interaction / s_interaction	Signed points	Raw pair payoff before level intensity and caps
categoryMix[].ruleId	String	Stable concentration rule identity
dominantThreshold	Share 0-1	Strict greater-than trigger
complementMinShare	Share 0-1	Strict less-than combined-complement trigger
r_mix / s_mix	Signed points	Triggered concentration effect

Event fields

Field	Meaning
id / scenarioId / deck	Stable event, scenario short ID, intermediate or final deck
weight	Relative draw weight within the eligible deck
immediateEffects	Raw signed risk, resilience, continuity, acceptability and budget vector
nextTurnEffects	Delayed signed vector; final-event delay is informational
mitigationRules[].tags	Any matching tag can activate this mitigation rule
riskReduction / resilienceProtection / continuityReduction	Raw mitigation magnitudes before capability scaling
amplificationRules[].missingTags	Amplification requires all listed tags to be absent
compatibilityTags	Tags for event protection and following-turn response coherence
exclusionTags	Retained metadata; current draw applies scenario/deck/non-repetition filtering
choices[]	Required revealed choice branches where present
choices[].titleKey / bodyKey	Keys resolved by the British-English content catalogue
durationTurns	Authored metadata; actual continuation is the explicit next-turn vector
sourceOrAssumption / auditRationale	Modelling provenance, not evidence of observed climate impacts

Run, snapshot and score fields

Field	Interpretation
phase	planning, event-reveal, event-choice, event or complete
turn	1, 2 or 3; each is an academic year
seed / prngState / prngVersion	Recorded deterministic draw identity and state
metrics	Current risk, resilience, continuity and acceptability

Field	Interpretation
active	Lifecycle records keyed by intervention ID; includes activating/damaged items
effectiveLevel / targetLevel	Current physical level and pending target
maturitySegments	Incremental start/upgrade segments and their saved delivery factors
pendingActivation	Delayed segment not yet physically active
damage / postRepairState	Event damage and the state restored by repair
budget.available	Opening planning resource authority
budget.carryOut	Remaining resource after commitment/event; displayed outside planning
budget.pendingDelta / eventDelta	Delayed opening change / current event resource change
coordination.netLoad / capacity / excessLoad	Turn-local delivery pressure and allowance
coordination.effectiveness	Positive operation increment coefficient
effects.activeCapacity.tagCapacity	Pre-event capability map, also used for response coherence
event.protection	Bound protection fixed for the source event
metricsBeforeEvent / metricsAfterEvent	Isolate immediate event resolution
event.eventOnlyMetricDeltas	Realised event-only changes after metric bounds
event.metricDeltas	Whole-turn difference from pre-commitment metrics; not event-only
history	Three post-event snapshots used by final scoring
pendingEffects	Delayed vectors scheduled for later application
continuingEffects	Informational effects outside the scored horizon
components.resilience	Mean of normalised cumulative and final resilience improvement
components.cumulativeRisk	Normalised mean-risk improvement relative to initial risk and floor
evaluationAdjustments	Diagnostics, not additional deductions applied after the composite

Verification and workbook boundary

Check parameter equality before interpreting any numerical example: all four release sums, all level tuples, six concentration rows and 272 pairs should match the canonical extract. Check that authored event counts are eight intermediate plus four final per scenario and that the eight opportunity events each expose their explicit choices. Empty operations remain a legal deliberate decision when the budget is not negative; they do not bypass event resolution.

For a formula workbook, disclose exactly which equations are implemented, their inputs, rounding and limits. A parameter table and isolated risk, coordination or scoring example do not reproduce lifecycle transitions, maturity-segment history, capacity aggregation, event draws, all choices, damage or compatibility validation. Formula cells must be recalculated and independently compared with expected values. Do not import the formula reference workbook as a player result archive: the game rejects formula-bearing result imports.

G2 author examples use deterministic synthetic decisions and current engine output. They demonstrate selected arithmetic and causal records, not strategy optimality, statistical balance, complete workbook parity or learning effectiveness. The author’s source checks and examples

require independent review before a G2 PASS. The final handoff records actual completed numerical and visual checks and any limitations; no check is implied merely because a document is readable.

Source lineage and supersession

The approved original DOCX framework, scenarios, intervention catalogue, payoff manual and XLSX supplied the initial identities and payoff structure. Their originals remain unchanged. V1 app calibration, V2.2 count-based overload and lifecycle rules, and earlier release-specific documents retain historical status. Current overrides are coordination-load-v2, Alpha-4 lifecycle simplification and Alpha-5 active-score-rebalance-v1, subsequently frozen in model V2-2.3.0. The earlier G1 run02 software qualification for 2.5.0 carries an accepted loss of 226 historical files; it is not an assertion of complete historical archive integrity.

The exact editable model is private maintenance material. The public technical tables and extract support inspection without exposing the entire development history. File disposition and final hashes are controlled by the dossier inventory and G2 handoff.

Coordination correction and exact rule identity

Owner decision DEC-G2-20260926-001 authorises canonical interaction lookup by the sorted intervention IDs. The earlier implementation missed 107 of 272 authored pair keys because their stored order differed from the lookup order. Both positive connectivity and conflict diagnostics now use all applicable authored records; pair payoff coefficients, model values and event data are unchanged. Multi-operation effectiveness and downstream results can change. This is not a V1 numerical model.

Current rule revision: `engine-v2.3.0-active-score-rebalance-coordination-v2`. Compatibility fingerprint: `rules-v2.3.0-coordination-pair-normalisation-2026-09-26.1`. Application version 2.5.0 and model version V2-2.3.0 alone do not identify these rules. The separate G1 run03 handoff records qualification results and the exact tested files for the corrected implementation. Earlier saves with the old rule fingerprint are rejected. No automatic migration or inherited statistical acceptance is claimed.